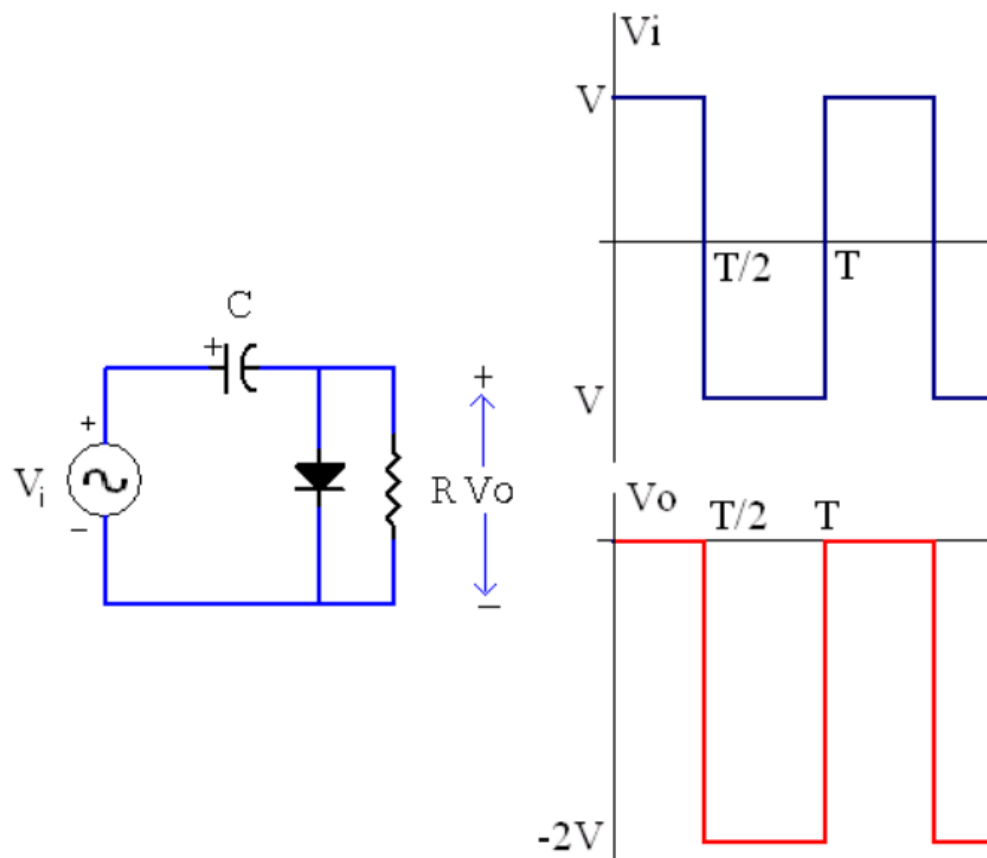


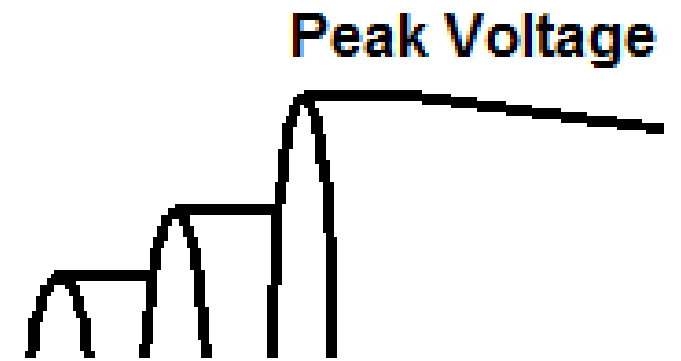
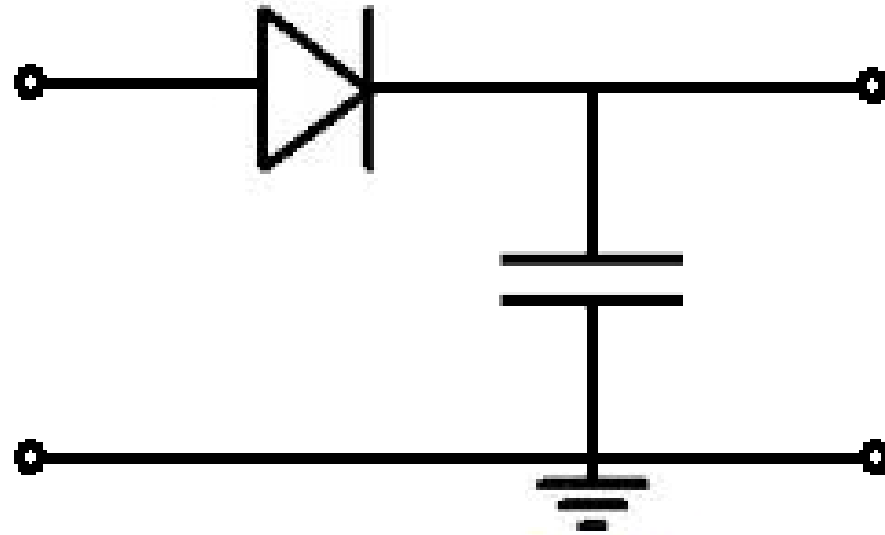
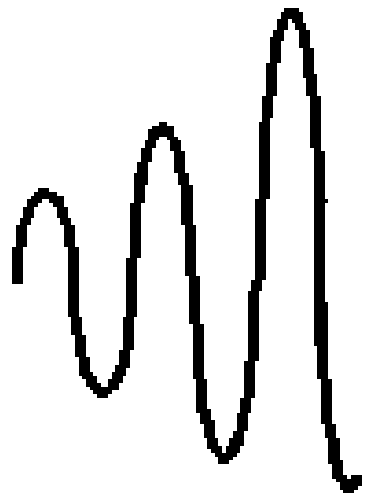
Clampers

A clamper is a network constructed of a diode, a resistor and a capacitor that shifts a **periodic** waveform to a different DC level without changing the appearance of the applied signal.

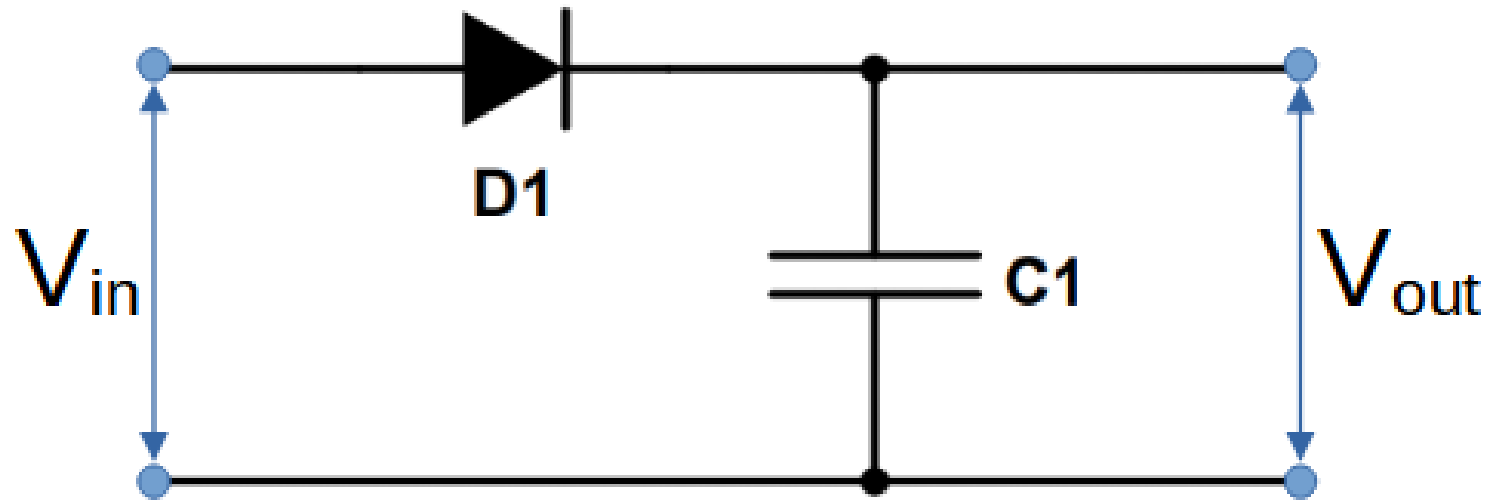
Example



Peak Detector



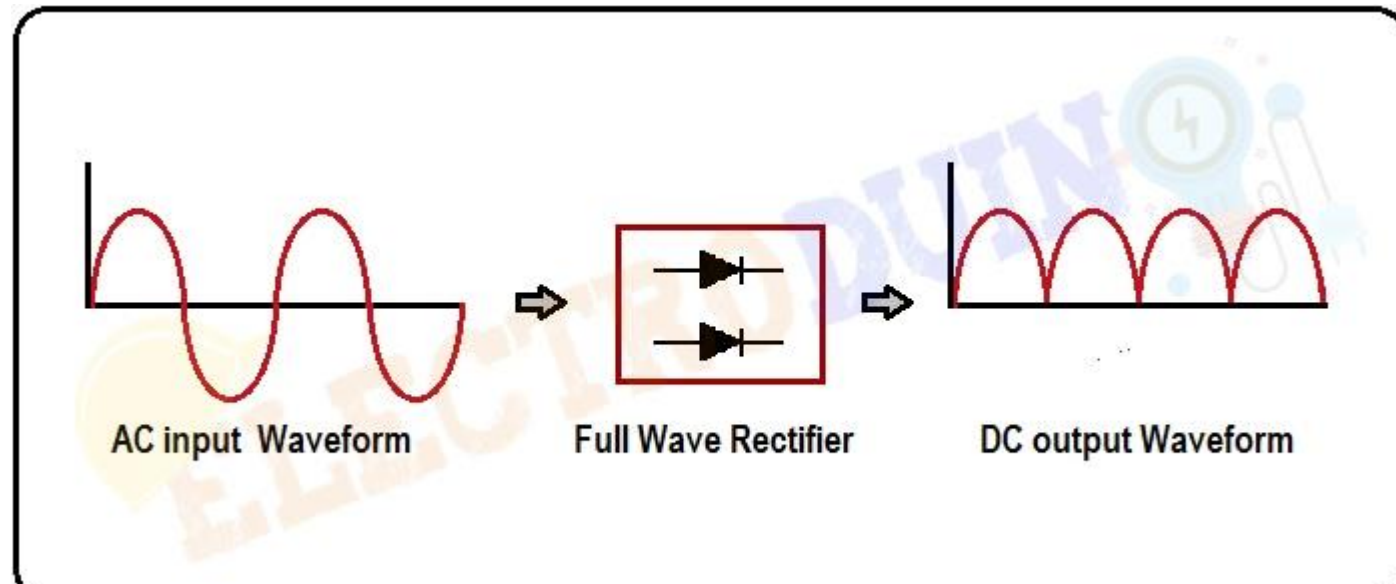
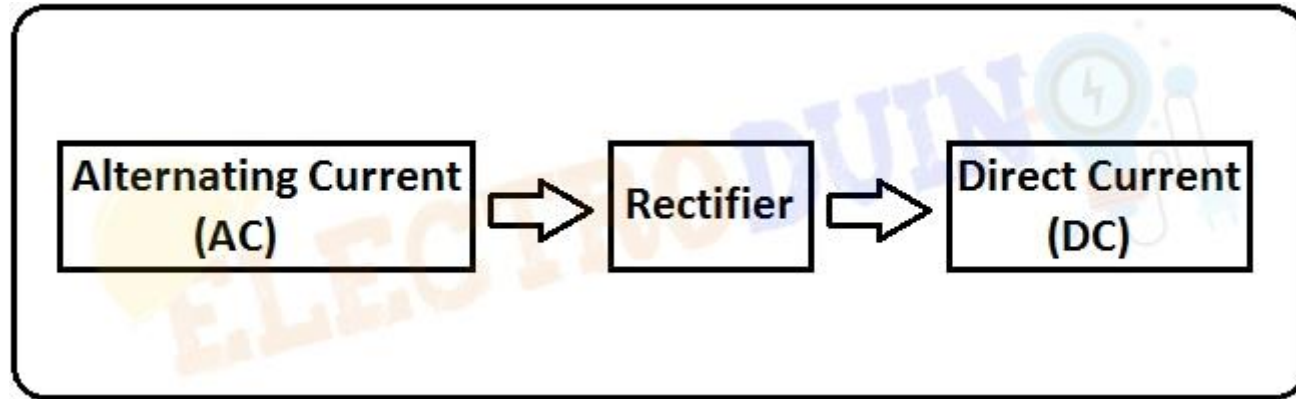
Peak Detector



In the positive half cycle, diode $D1$ is forward biased and capacitor $C1$ starts charging. When input reaches its peak value capacitor gets charged to positive peak value.

In negative half cycle, as input decreases, diode $D1$ is reversed biased and capacitor is isolated and holds the peak value of previous cycle. Hence called as peak detector.

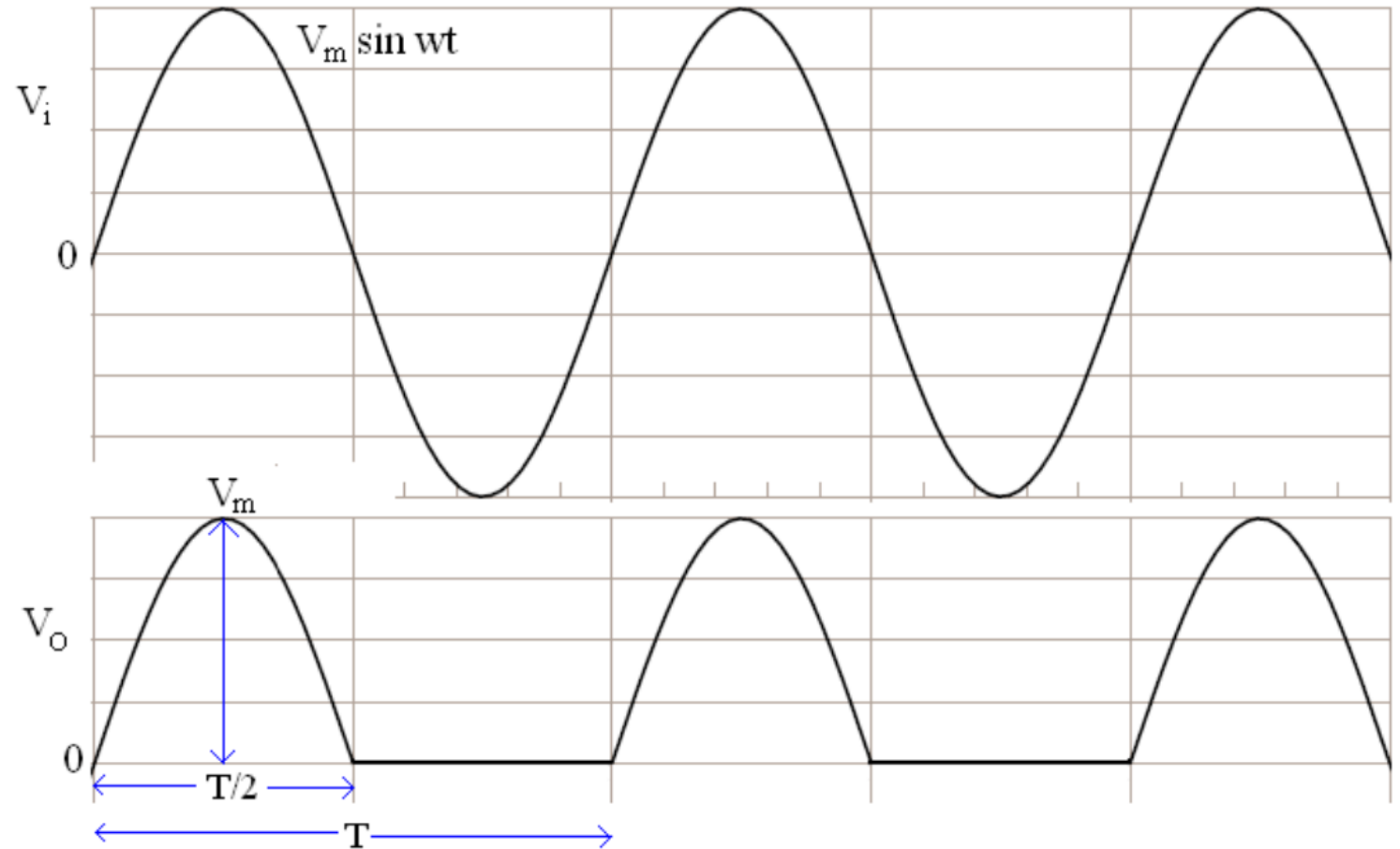
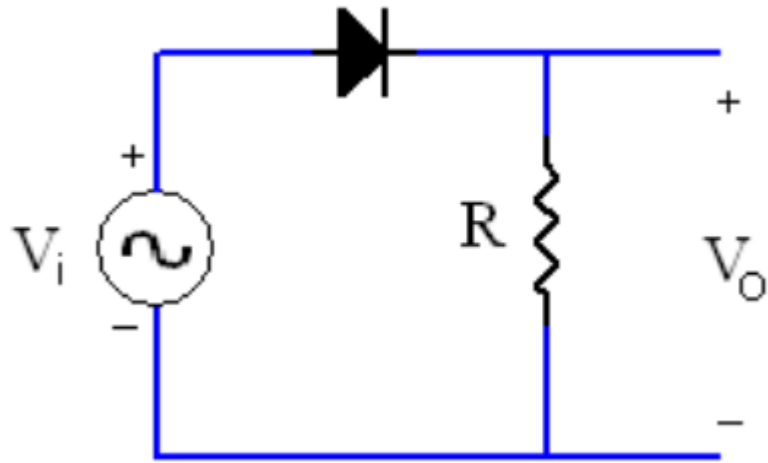
Rectifier Circuits



Rectifier Circuits

Half-wave rectifier (HWR)

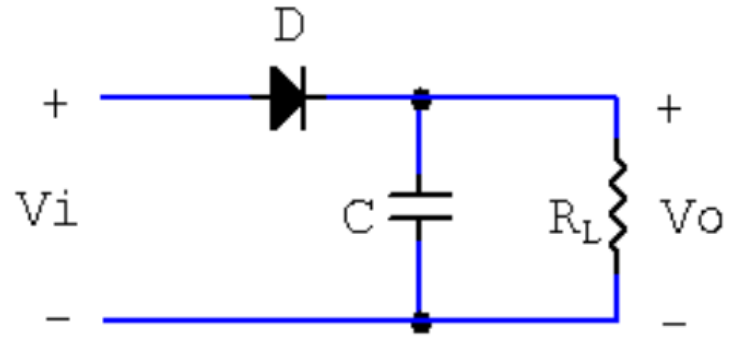
Considering the diode to be ideal



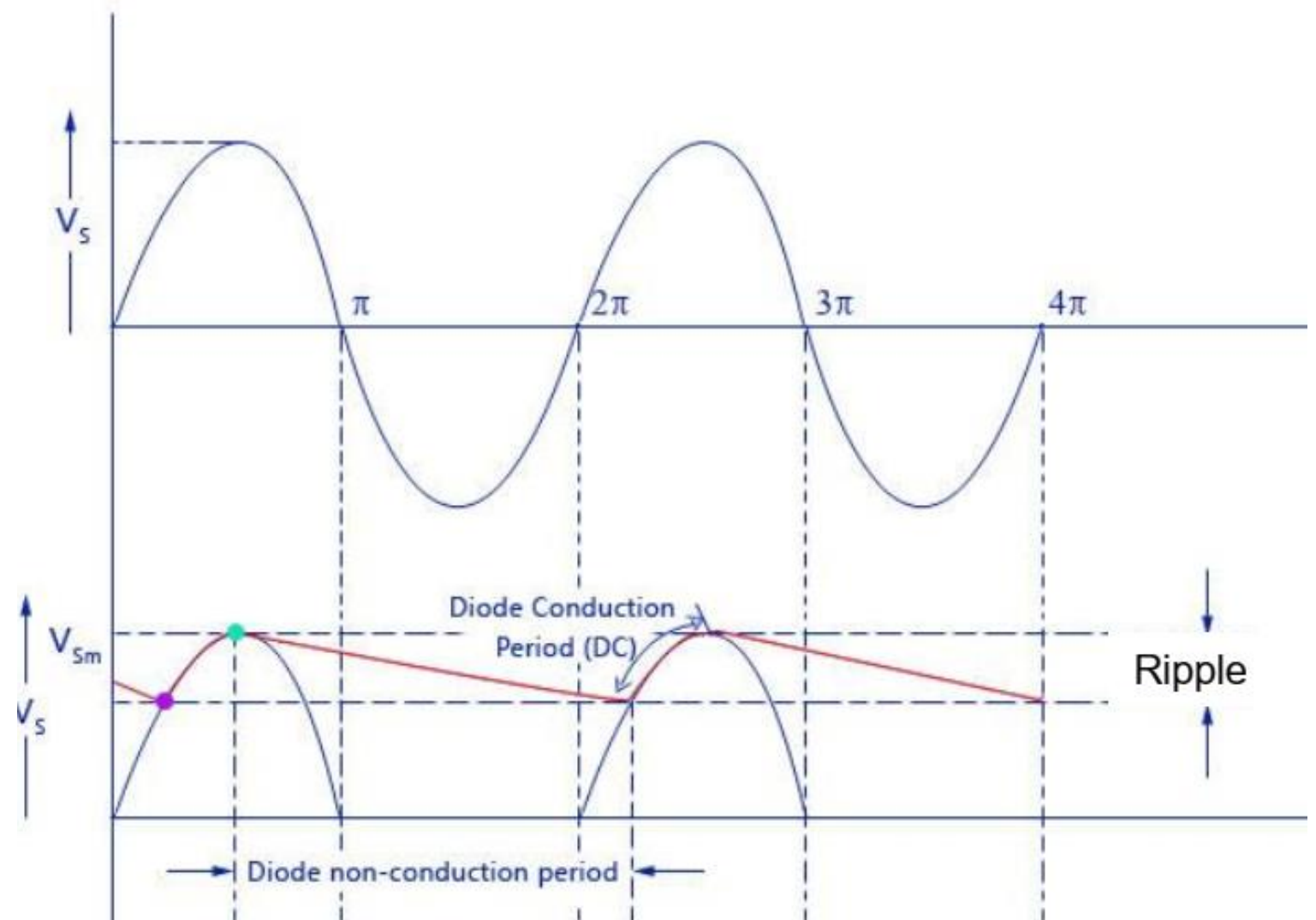
$$\begin{aligned}
 V_{\text{DC}} &= \frac{1}{T} \left[\int_0^{T/2} V_m \sin \omega t dt + \int_0^{T/2} (0) dt \right] & \frac{2\pi}{\omega} = T \\
 &= \frac{-V_m}{\omega T} \left[\cos \omega t \right]_0^{T/2} \\
 &= \frac{V_m}{\pi}
 \end{aligned}$$

$$V_{\text{DC}} = 0.318 V_m$$

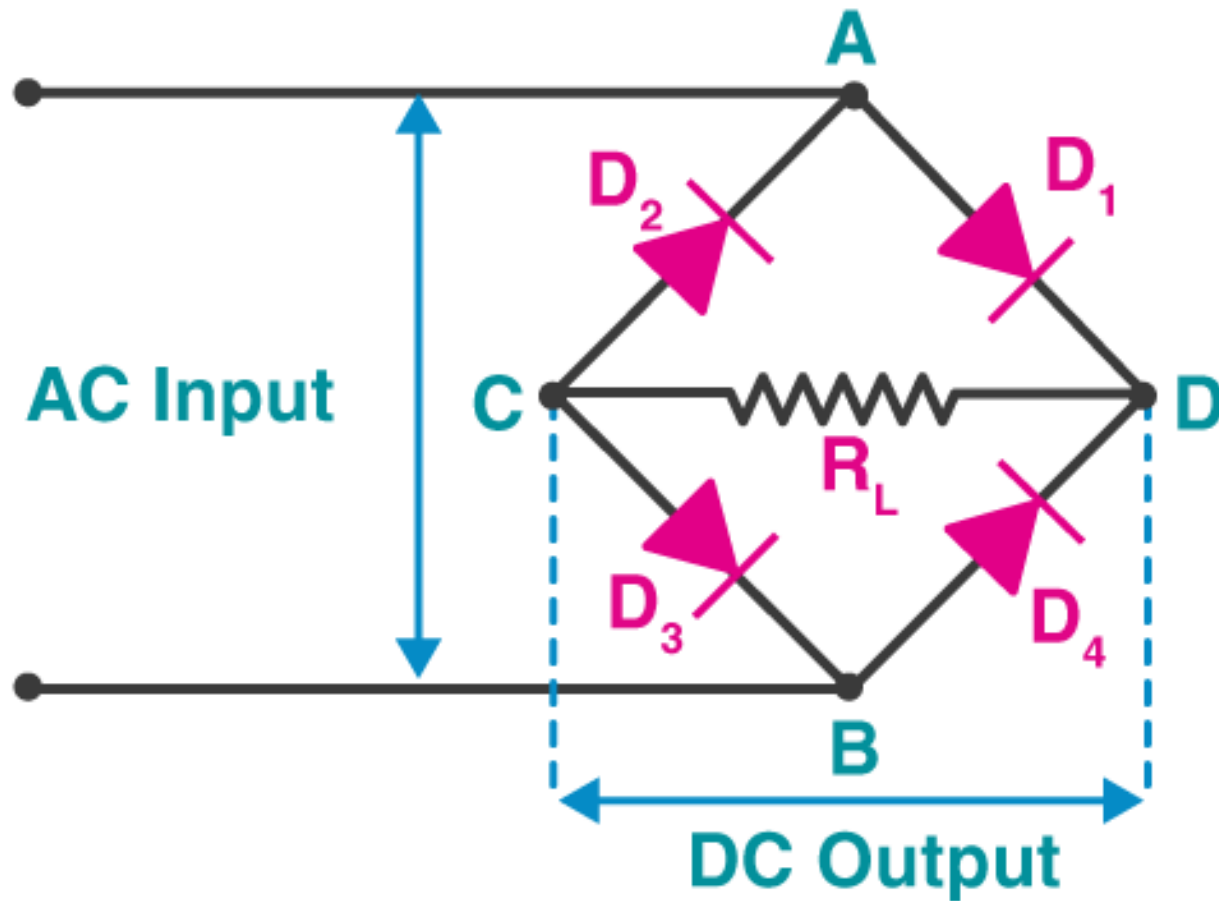
HWR with Capacitor Filter

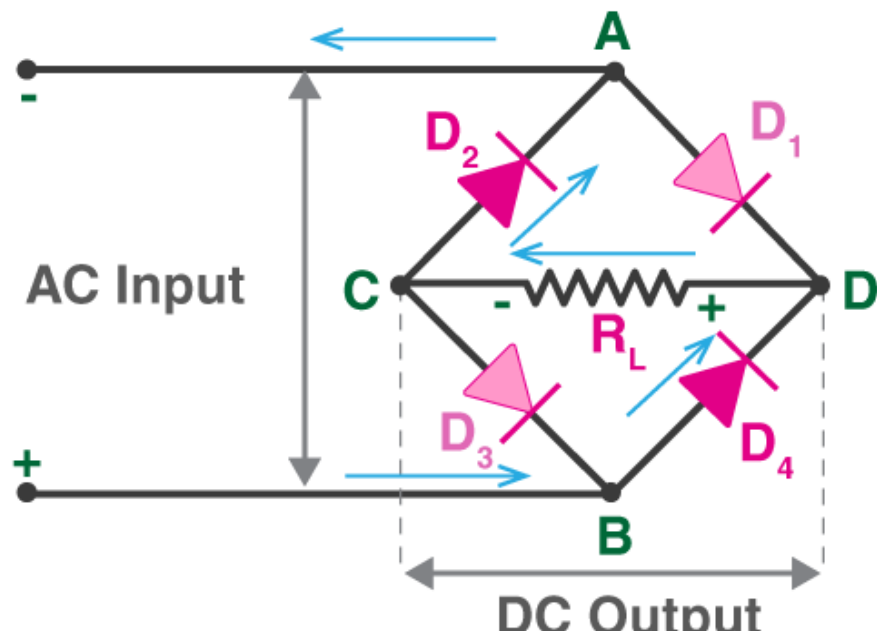
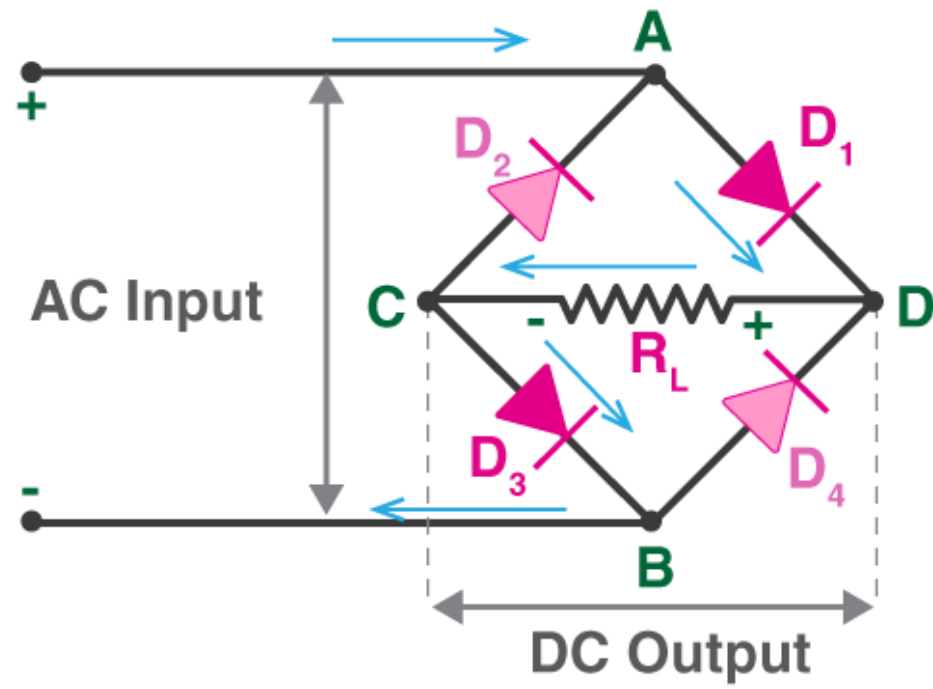


Assuming ideal diode
for simplicity,
 $V_D = 0$, $R_f = 0$



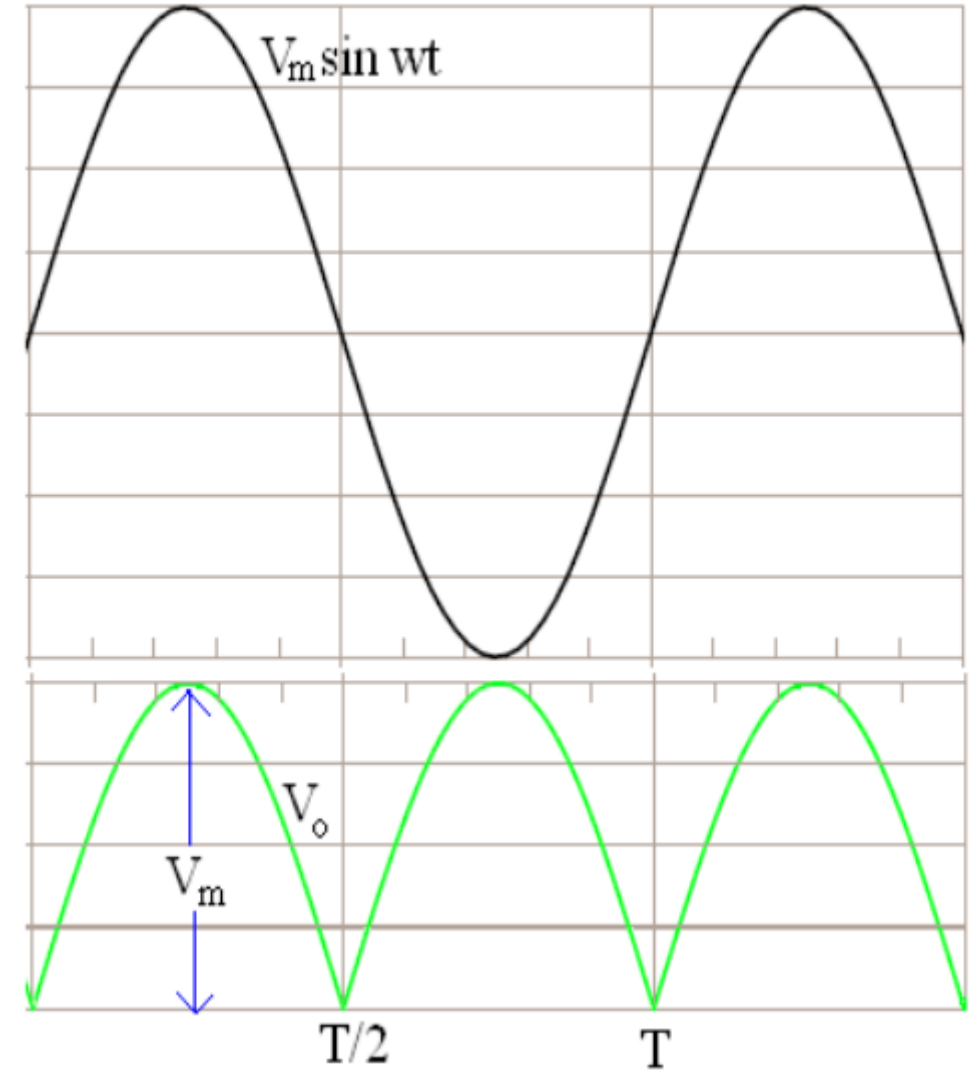
Bridge Full Wave Rectifier

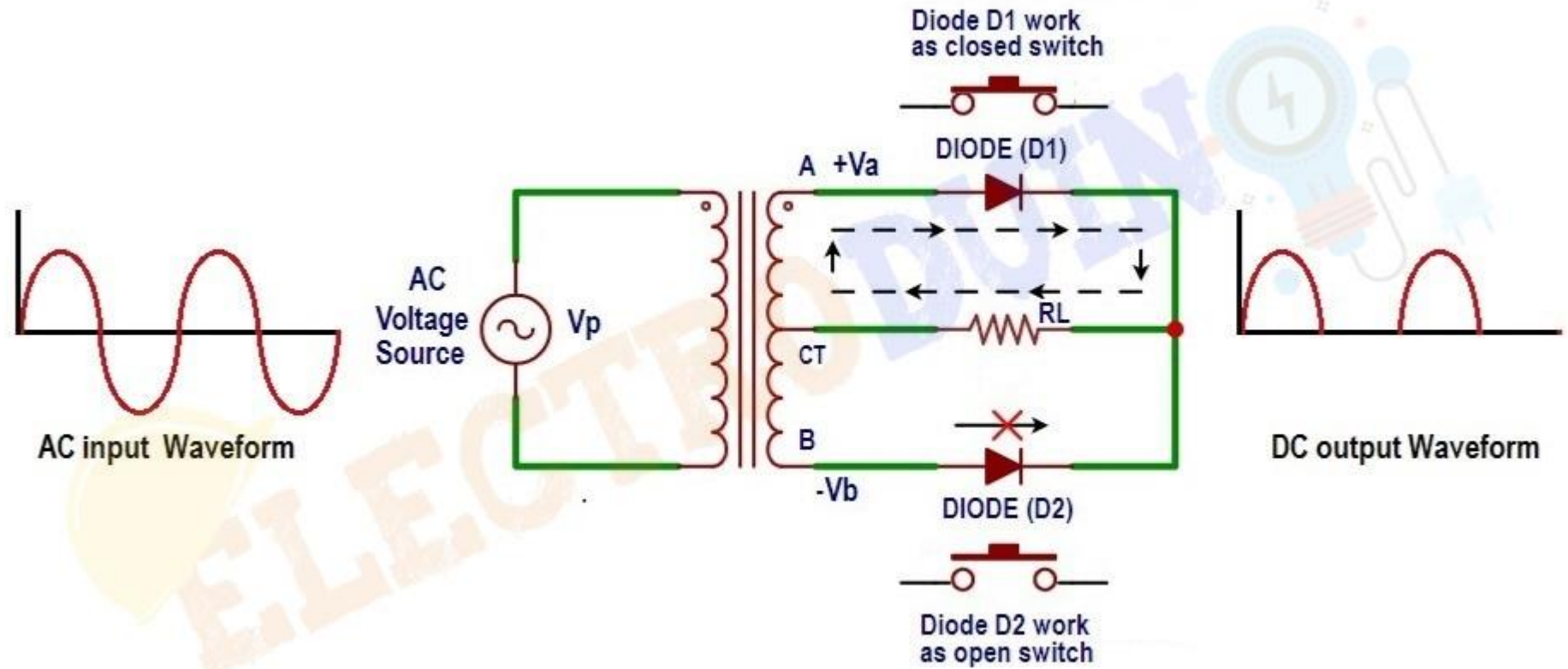


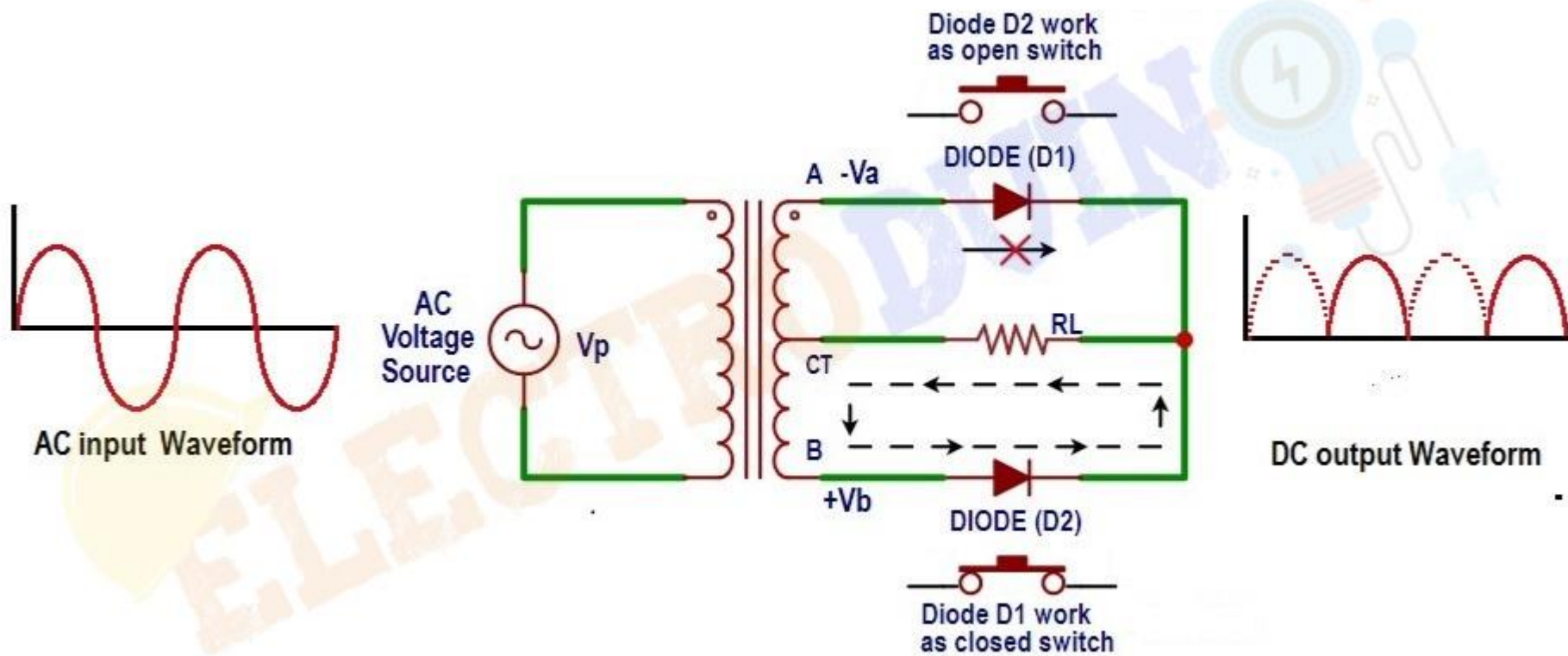


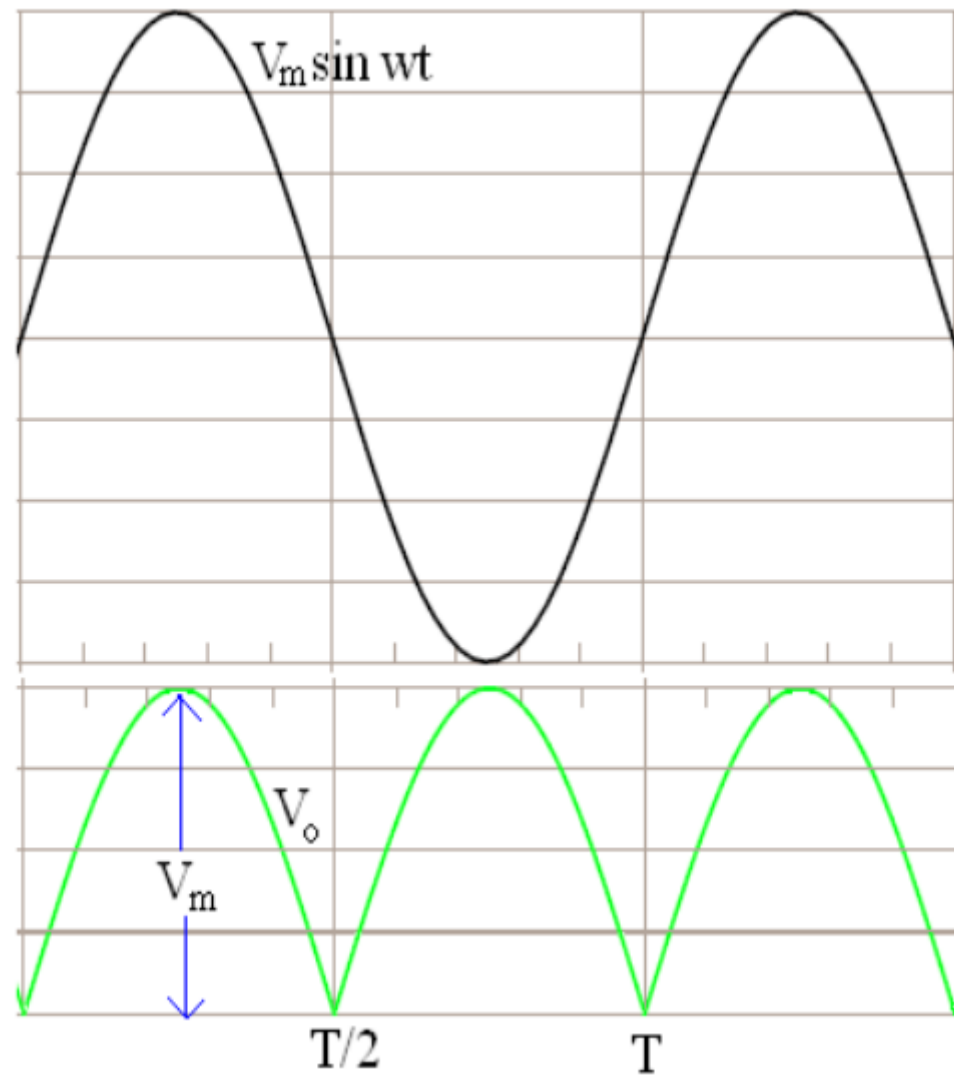
Assuming ideal diode,

$$\begin{aligned} V_{\text{DC}} &= \frac{2}{T} \int_0^{T/2} V_m \sin \omega t dt \\ &= \frac{-2V_m}{\omega T} [\cos \omega t]_0^{T/2} \\ &= \frac{2V_m}{\pi} \end{aligned}$$

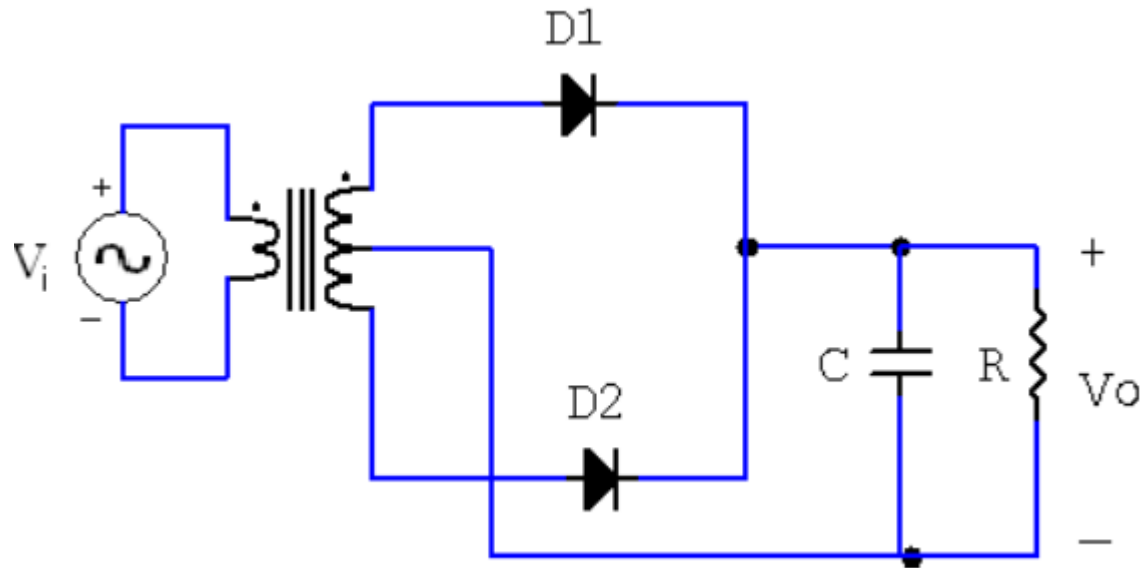




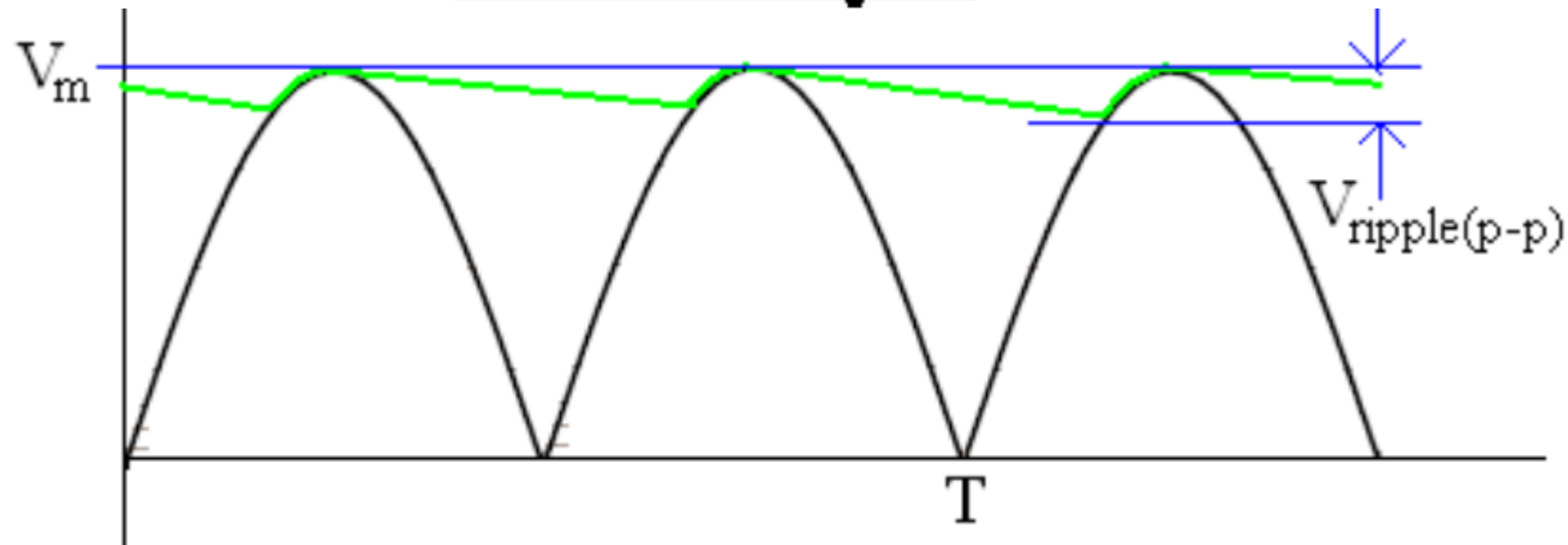


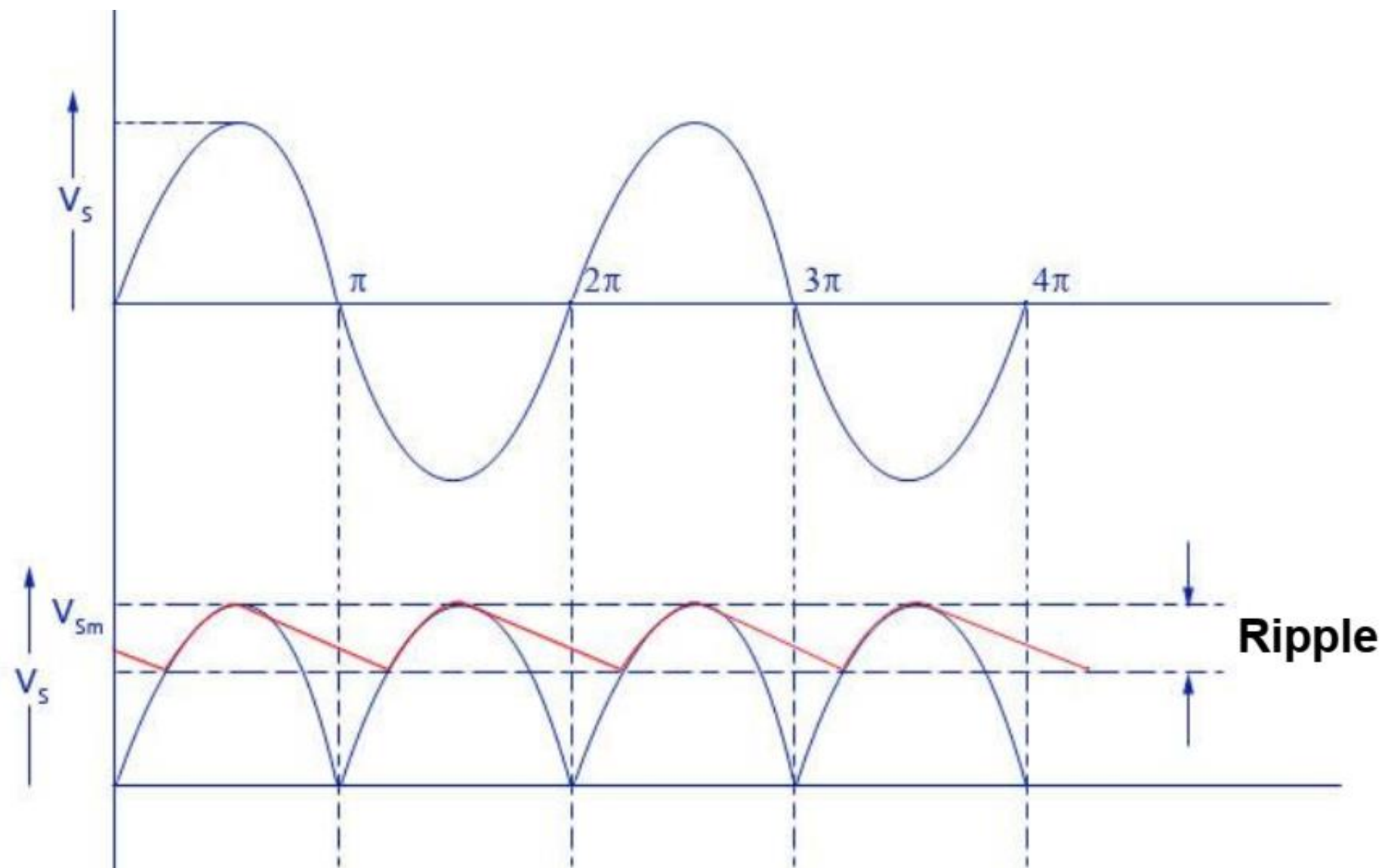


FWR with Capacitor Filter



Assuming triangular ripple waveform,
 $V_o(\text{dc}) = V_m - V_{r(\text{p-p})}/2$

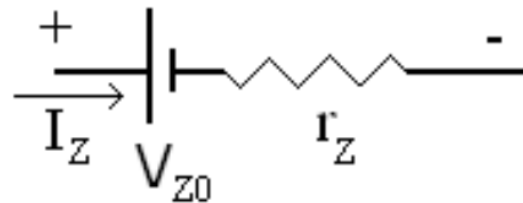
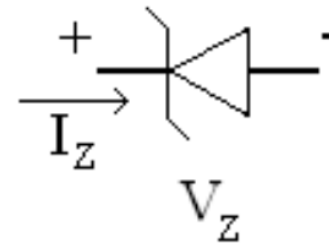




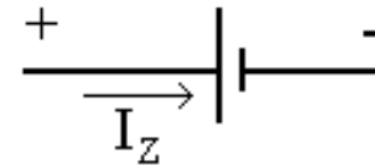
Zener Diode

Zener diodes are special diodes manufactured with adequate power dissipation capabilities to operate in the breakdown region.

Symbol:



$$V_Z = V_{Z0} + I_Z r_Z$$

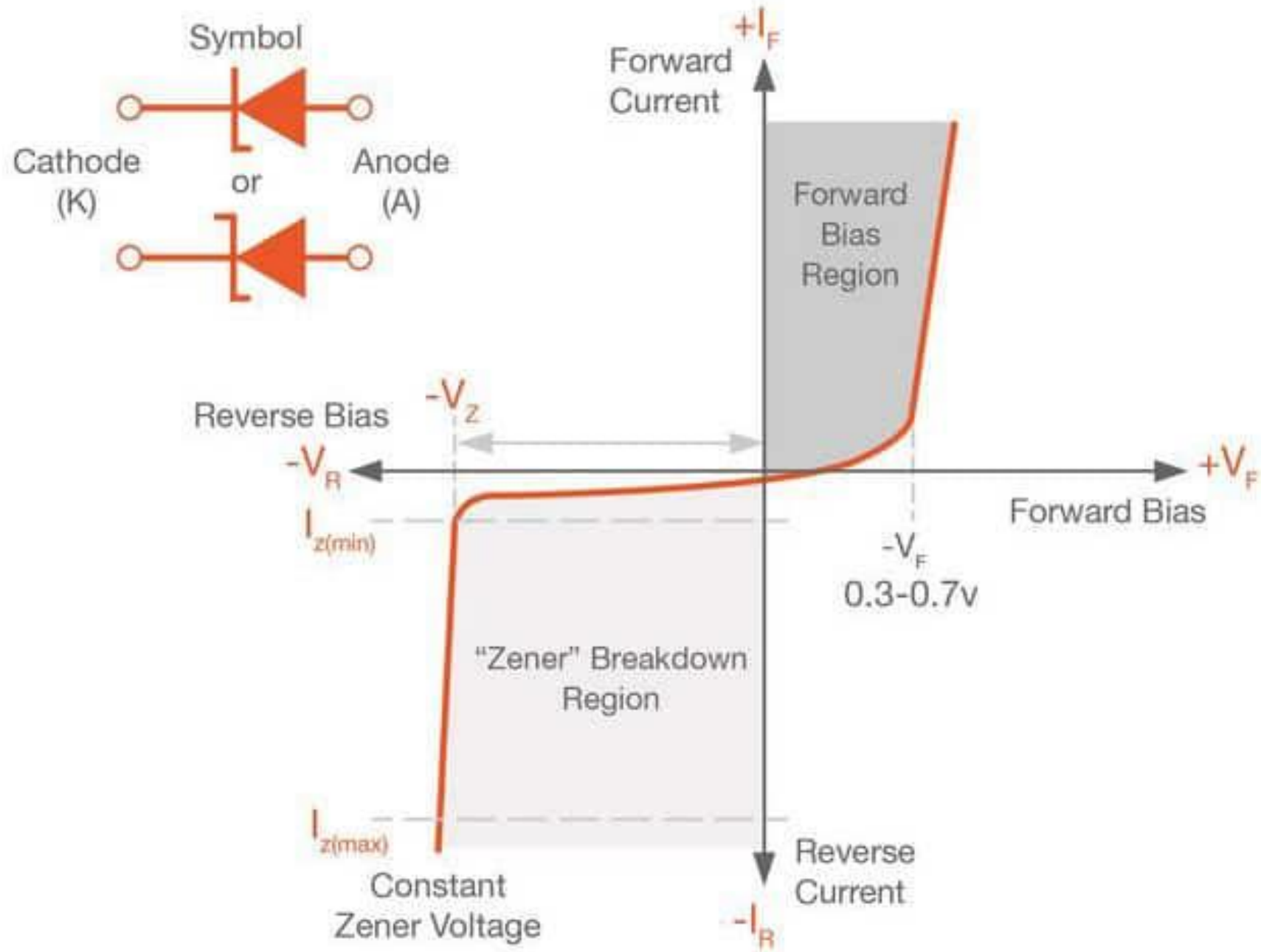


$$V_Z = V_{Z0}$$

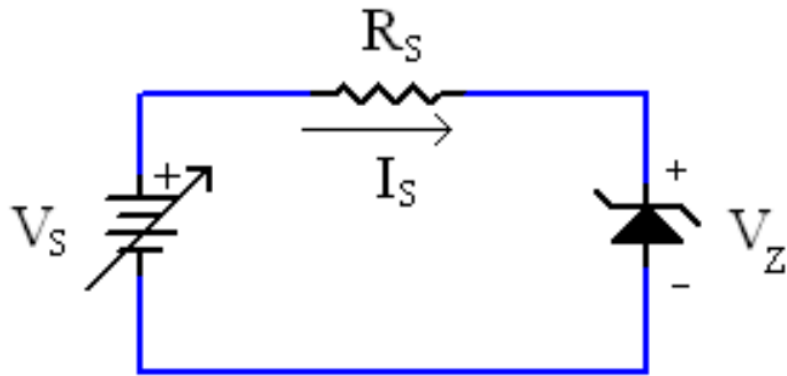
($I_Z r_Z$ is ignored)

Equivalent ckt.

Approx. Eq. ckt.



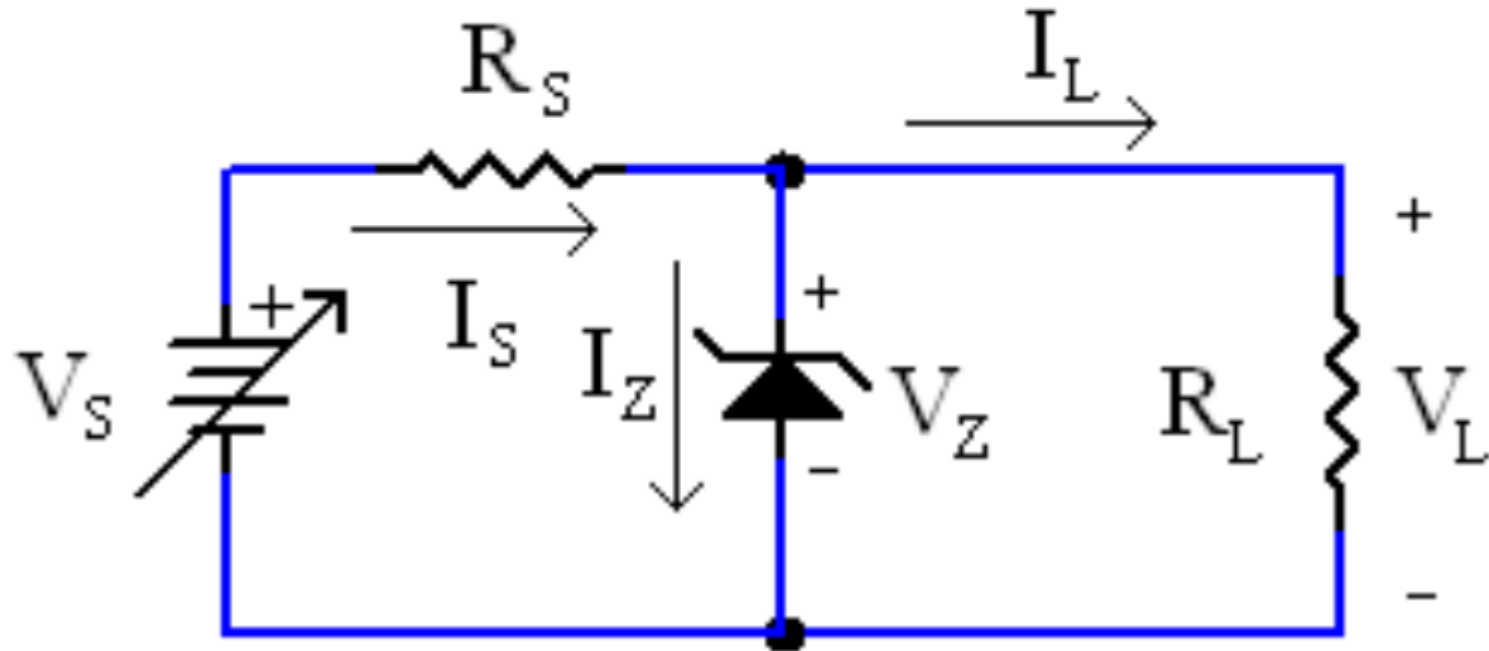
Zener Regulator



- $V_s > V_z$ for breakdown
- R_s is the current limiting resistance to limit the zener current to less than its maximum rating

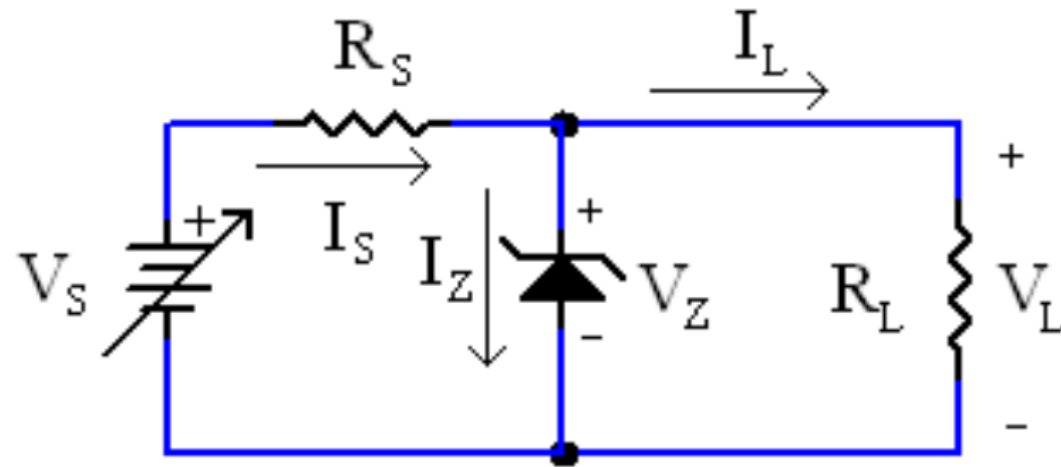
$$I_s = (V_s - V_z) / R_s$$

Loaded Zener regulator



V_{th} = voltage across zener when
it is not in breakdown
 $= V_S \times R_L / (R_S + R_L)$

Loaded Zener regulator



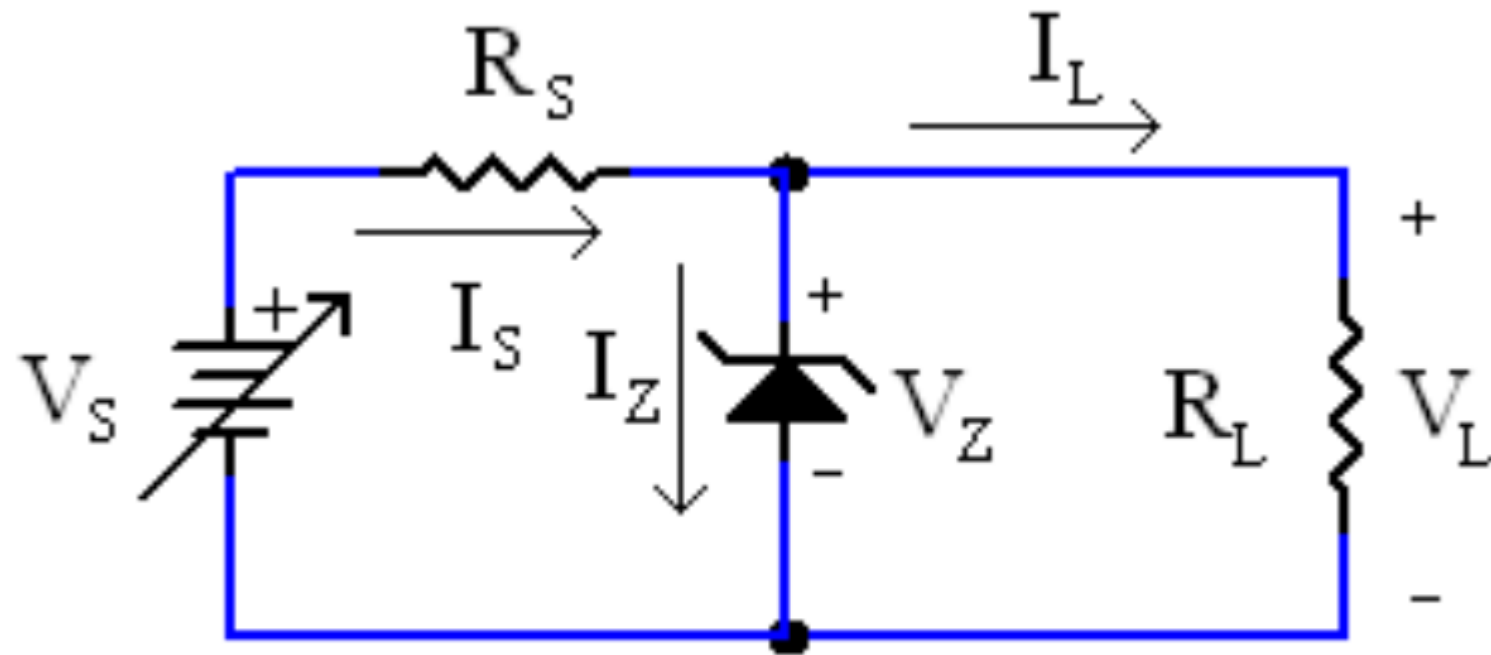
For breakdown, $V_{th} > V_Z$

$$I_L = V_L / R_L = V_Z / R_L$$

$$I_Z = I_S - I_L$$

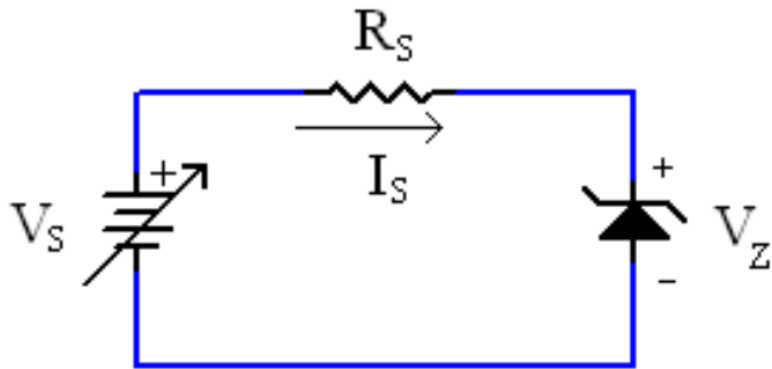
Power dissipated by
zener diode = $V_Z I_Z$

Loaded Zener regulator



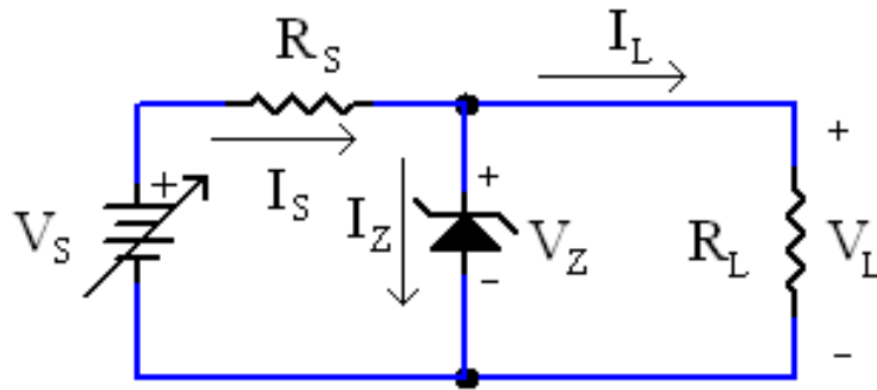
$$I_S = (V_S - V_Z) / R_S, \quad I_S = I_Z + I_L$$

Zener Regulator



Typically, for a Zener Diode, one would specify the **zener voltage** V_Z and the **maximum power dissipation** $P_{Z,max}$ in the zener diode.

Loaded Zener regulator



$$I_S = (V_S - V_Z) / R_S, \quad I_S = I_Z + I_L$$

Additionally, we may also specify the **minimum zener current** $I_{Z,\min}$ that must flow through the zener diode to provide the zener action